THE PLAYLIST LESSON · LESSON 2 OF 6

What’s in the Playlist?

A 45-minute lesson on how AI learns — and where bias comes from. The hook: an AI is a music app, and training data is its playlist. It can only mix what's in the playlist, so a one-genre playlist means one-genre answers, no matter who's asking.

AT A GLANCE

TIME	MATERIALS	TECH NEEDED	PREP
45 min	Playlist Check worksheet (1 per student; pairs share one request line; scissors optional)	None — fully unplugged	Print worksheets. That's it.

OBJECTIVES

- Explain how an AI learns: **input** → **patterns** → **output** — prediction from data, not magic.
- Identify who a dataset serves well, who it leaves out, and connect that gap to real AI bias.
- Discuss data gaps collaboratively: say back a partner's idea and build on it.

STANDARDS ALIGNMENT

CSTA 2-IC-21 | ISTE 1.1.d | CCSS ELA SL.6.1

Full standards text in the Teacher Guide.

WARM-UP — THE ONE-PLAYLIST APP (7 MIN)

Announce that you are **AutoMix**, the school's new music app. Your playlist holds one kind of song: loud, fast party tracks. Take 3–4 requests and serve every one with total confidence:

"Something chill to study to?" → "Here's 'TURBO TURBO.' It is not chill. It's what I've got."

"A song for grandma's birthday dinner?" → "Say less. 'VOLUME WAR.' Grandma loves bass drops, probably."

"A slow song for the last dance?" → "'NEON STAMPEDE,' played at... the exact same speed."

Then ask: **is AutoMix bad at picking songs — or is something else broken?** Fish for it: the app is doing its job perfectly. The *playlist* is the problem.

> **Bridge:** "If you did the genie lesson: it granted what you SAID. Today: where its answers even come from. Everything an AI 'knows' comes from a playlist of examples — its training data. It can remix the playlist, but it can't serve what was never in it. Today you'll BE the app."

DIRECT INSTRUCTION — NO MAGIC IN THE MACHINE (8 MIN)

Write the chain on the board: **INPUT → PATTERNS → OUTPUT.**

INPUT — the playlist. Training data: the giant pile of examples an AI studies before you ever meet it — songs for a music app, mountains of human writing for a chatbot. Nobody types "knowledge" in. It studies examples.

PATTERNS — the training. The AI works through the playlist and gets good at ONE thing: guessing what usually goes with what. It doesn't understand any of it — it finds patterns.

OUTPUT — the mix. Ask for something and it predicts what best matches its playlist's patterns. A prediction, not a lookup — which is why it can sound confident and still be off.

Say it plainly: "No magic in the machine. An AI doesn't KNOW things — it predicts things, from its playlist. So two rules run everything: a GAP in the playlist is a gap in the AI, and a LEAN in the playlist leans every answer. That lean has a name: bias."

Quick check: "A photo AI's playlist is almost all golden retrievers. A husky walks in — what's it going to call it?" (Its best guess from its patterns: golden retriever. Gap in, gap out.)

ACTIVITY — PLAYLIST CHECK (20 MIN)

1. **Setup (3 min):** Pairs — one partner is the **App**, one runs the **Request Line**. The 12 song cards ARE the app's training data. House rules: serve only from the playlist, and never refuse — serve the closest thing you've got.
2. **Serve the Request Line (9 min):** Work the six printed requests (Keisha, Nia, Tomás, Priya, Malik, Jordan), swapping roles after three. The App picks the best card; both rate the fit. End with one live request from a neighboring pair.
3. **Playlist Gap Report (5 min):** Pairs record who the playlist serves well and who it leaves out. Discussion rule, enforced out loud: *before you add your idea, say back your partner's and build on it.*
4. **Fix the Playlist (3 min):** Invent 3 new cards that fix the biggest gaps. Push past "add one slow song" — which requests still have nothing?

DEBRIEF — THE REAL PLAYLISTS (5 MIN)

Pairs call out one gap they found. Connect: "Your playlist failed Tomás and Priya. Real playlists fail real people the same way." Walk the three examples in the Teacher Guide, then land the big question: **"Nobody put bad songs in your playlist — so where did the bias come from?"** (From what got collected and what didn't.)

EXIT TICKET (5 MIN)

Bottom of worksheet: complete the **input** → **patterns** → **output** chain, then diagnose a lopsided homework-helper AI using the word "playlist."

DIFFERENTIATION

- **Support:** Sentence frames + pre-sorted playlist note (Teacher Guide, p.2).
- **Extension:** Design a 12-card playlist that serves every request on the sheet — then defend the trade-offs.
- **ELL:** Ratings are check-marks, not sentences; invented cards may be in any language (which fixes a real gap).
- **No-device classrooms:** Built in — the whole lesson is paper, pairs, and one loud imaginary music app.